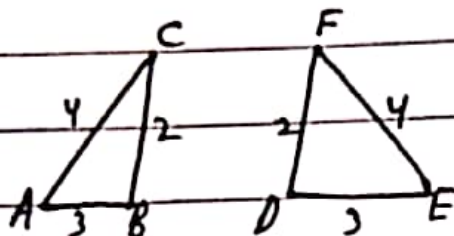
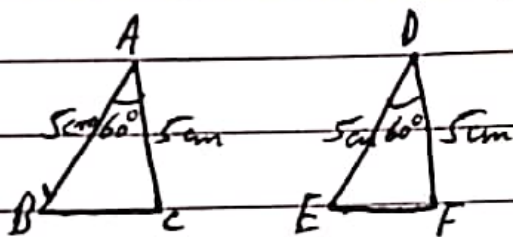


Circles.→ Rules of Congruence:→ Rule 1: SSS - (Side Side Side).

$$AB = DE = 3\text{cm}$$

$$AC = EF = 4\text{cm}$$

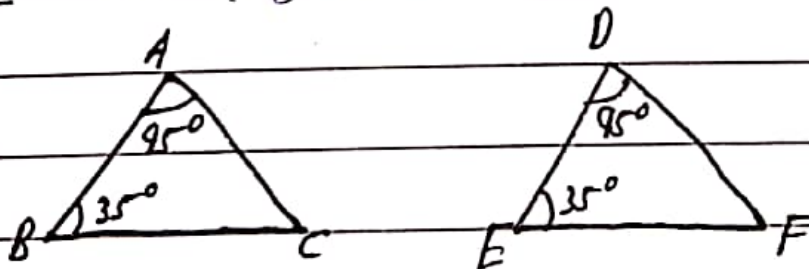
$$BC = DF = 2\text{cm}$$

 $\therefore \triangle ABC$ is congruent to $\triangle DEF$ ($\triangle ABC \cong \triangle DEF$).
→ Rule 2: SAS (Side Angle Side)

$$\angle A = \angle D = 60^\circ$$

$$AB = DE = 5\text{cm}$$

$$AC = DF = 5\text{cm}$$

 $\therefore \triangle ABC \cong \triangle DEF$
→ Rule-3: ASA (Angle side Angle).

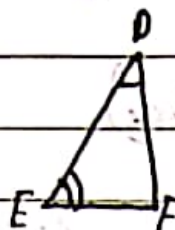
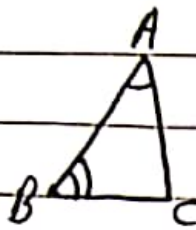
$$\angle A = \angle D = 95^\circ$$

$$\angle B = \angle E = 35^\circ$$

$$AB = DE$$

 $\therefore \triangle ABC \cong \triangle DEF$

→ Rule 4: AAS (Angle Angle Side)



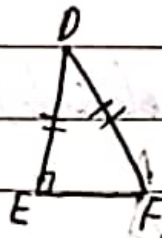
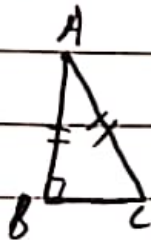
$$\angle A = \angle D$$

$$\angle B = \angle E$$

$$BC = EF$$

$$\triangle ABC \cong \triangle DEF$$

→ RHS Rule: (Right Hypotenuse Side)



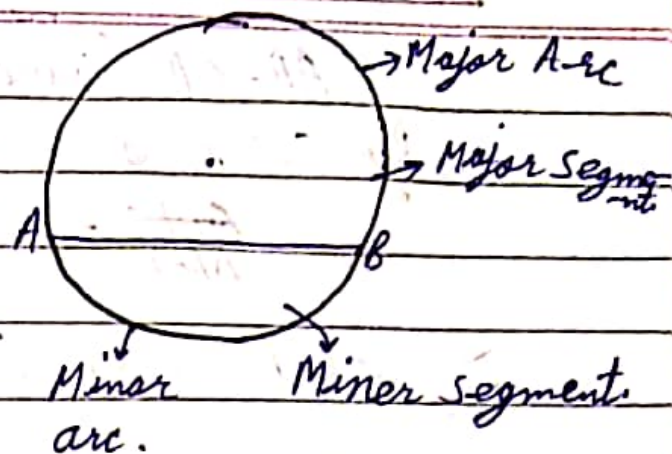
$$AC = DF$$

$$AB = DE$$

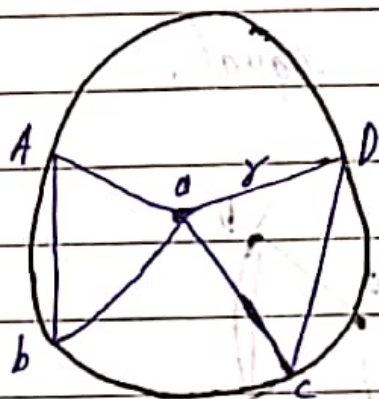
$$\triangle ABC \cong \triangle DEF$$

⇒ Circle - Circle is the infinite set of points equidistant from a fixed point.

→ Chord - A line segment with its end points on the boundary of the circle.



- Theorem 9.1: Equal chords of a circle subtend equal angles at the centre.



- Given: Circle with centre O and radius r , i.e. $C(O, r)$.
Chords AB and CD are such that:
 $AB = CD$.

- To Prove: $\angle AOB = \angle COD$

- Proof: In $\triangle AOB$ and $\triangle COD$

$AO = OD$ (Radii of same circle)

$OB = OC$ (Radii of same circle)

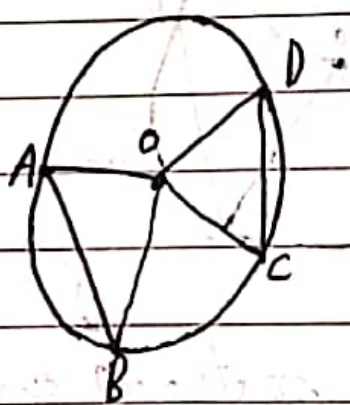
$AB = CD$ (Given)

By SSS congruence

$\triangle AOB \cong \triangle COD$

$\therefore \angle AOB = \angle COD$ (C.P.C.T)

$\Rightarrow$ Theorem 9.2 - If the Angles subtended by the chords of a circle at the centre are equal, then the chords are equal.



$\rightarrow$ Given: A circle with centre O, chords AB and CD, such that,

$\angle AOB = \angle COD$

$\rightarrow$ To Prove - $AB = CD$

$\rightarrow$ Proof - In $\triangle AOB$ and $\triangle COD$.

$AO = OD$ (Radii of the same circle).

$OB = OC$ (Radii of the same circle).

$\angle AOB = \angle COD$ (Given)

$\therefore$ By SAS congruence.

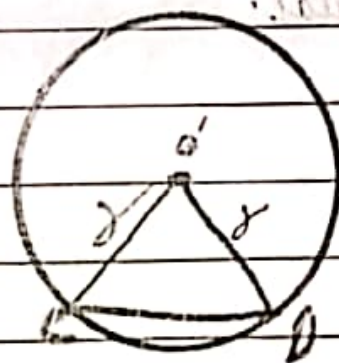
$\triangle AOB \cong \triangle COD$

$AB = CD$ (C.P.C.T)

EXERCISE - 8.1

Q1) Recall that two circles are congruent if they have the same radii.?

Sol.:



Given: Two Circles

$C_1 (O, r)$ and

$C_2 (O', r)$

- Chords AB of C_1 and CD of C_2 are, such that.
 $AB = CD$.

→ To prove - $\angle AOB = \angle CO'D$

- Proof - In $\triangle AOB$ and $\triangle CO'D$
 $AO = CO' = r$ (Circles are congruent)
 $BO = DO = r$ (Circles are congruent)
 $AB = CD$ (Given)

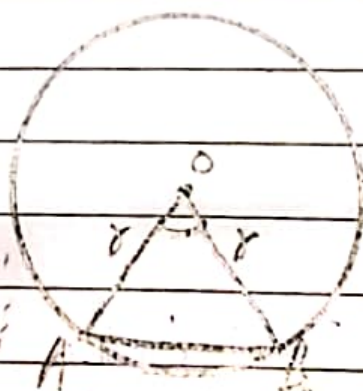
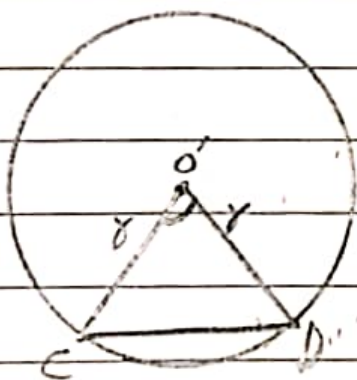
$\therefore$ By SSS Congruence

$$\triangle AOB \cong \triangle CO'D$$

$$\angle AOB = \angle CO'D \text{ (C.P.C.T.)}$$

Q2) Prove that if chords of congruent circles subtend equal angles at their centres, then the chords are equal.

Sol:



- Given - Two circles.

Circle 1 (O', r)

Circle 2 (O, r)

$$\therefore \angle CO'D = \angle AOB$$

- To prove: $CD = AB$

- Proof: In $\triangle AOB$ and $\triangle CO'D$

$$\angle CO'D = \angle AOB \text{ (Given)}$$

$$CO' = AO = r$$

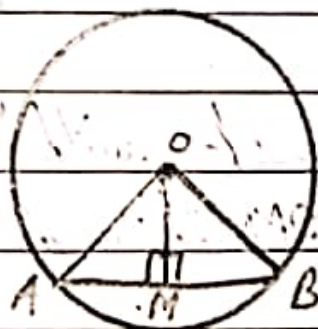
$$BO = DO' = r$$

$\Rightarrow$ By ~~SAS~~ SAS Congruence

$$\triangle AOB \cong \triangle CO'D$$

$$AB = CD \text{ (C.P.C.T)}$$

$\Rightarrow$ Theorem 9.3 - The Perpendicular from the centre of a circle to a chord bisects the chord.

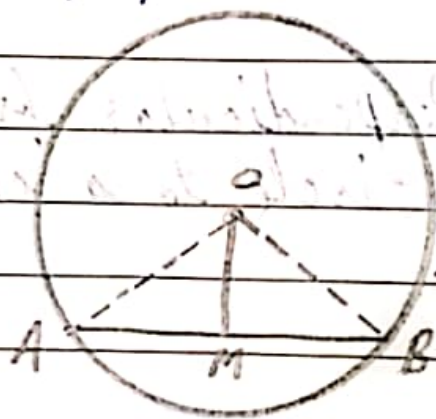


- Given: A circle with centre O, Chord AB and $OM \perp AB$.

- To prove: $AM = MB$

- Const. Join OA and OB
- Proof. In $\triangle AMO$ and $\triangle BMO$
 $OA = OB$ (Radii of same circle)
 $OM = OM$ (Common)
 $\angle AMO = \angle BMO$ (Each 90°)
 $\Rightarrow$ By SSA Congruence
 $\triangle AMO \cong \triangle BMO$
 $AM = BM$ (C.P.C.T)

$\Rightarrow$ Theorem (9.4). A line drawn from the centre of the circle to bisect the chord is perpendicular to the chord.



- Given. A circle with centre O , AB is chord and line OM through centre ' O ', such that $AM = MB$
- To prove. $OM \perp AB$
- Const. Join AO and BO .

- Proof- In $\triangle AMO$ and $\triangle BMO$
 $AO = BO$ (Radii of same circle)
 $OM = OM$ (Common)
 $AM = MB$ (Given)

- By SSS congruence

$$\triangle AMO \cong \triangle BMO$$

$$\angle AMO = \angle BMO = x.$$

- Now, $\angle AMO + \angle BMO = 180^\circ$

$$x + x = 180^\circ$$

$$2x = 180^\circ$$

$$x = \frac{180^\circ}{2} = 90^\circ$$

$$\therefore \angle AMO = \angle BMO = x = 90^\circ$$

$$\Rightarrow OM \perp AB.$$

$\Rightarrow$ Theorem (9.5) Equal chords of circle (or congruent circle) are equidistant from the centre.



- Given- A circle with centre 'O'. AB and CD are

two chords of the same circle

$$AB = CD$$

$$OM \perp AB$$

$$ON \perp CD$$

- To prove $OM = ON$

- Constr. Join AO and OB .

- Proof: Since $OM \perp AB \Rightarrow AM = MB$ --- ①

$$ON \perp CD \Rightarrow CN = ND$$
 --- ②

$$AB = CD \text{ (Given)}$$

$$AM + MB = CN + ND$$

$$AM + AM = ND + ND$$

$$2AM = 2ND$$

$$\boxed{AM = ND}$$
 --- ③

- In $\triangle AMO$ and $\triangle DNO$

$$AO = DO \text{ (Equal Radii)}$$

$$AM = ND \text{ (Proved)}$$

$\therefore$ By RHS congruence

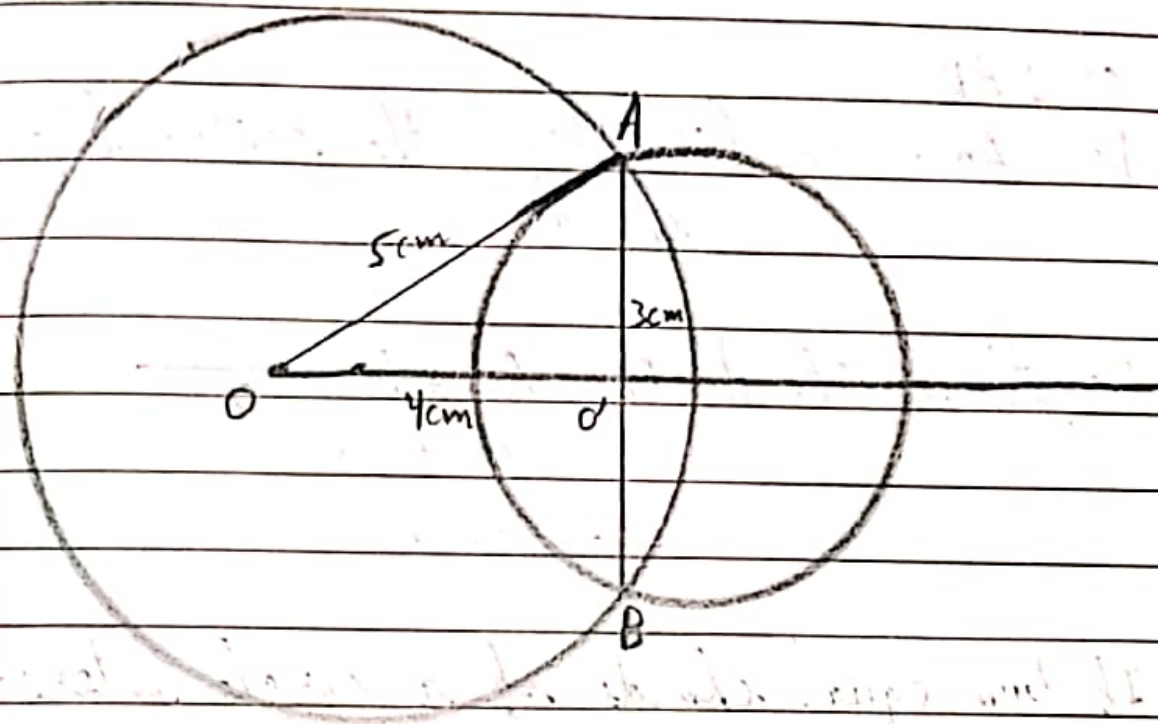
$$\triangle AMO \cong \triangle DNO$$

$$\Rightarrow OM = ON \text{ (C.P.C.T.)}$$

Exercise = 8.2

Q1:- Two circles of radii 5cm and 3cm intersect at two points and the distance between their centres is 4cm. Find the length of common chord.

Soln



- Given: Two circles with centre O and O',
Radii 5cm and 3cm. Such that,
 $OO' = 4\text{cm}$

- Const- Join OA.

- In $\triangle OO'A$

$$(OA)^2 = 5^2 = 25 \text{ --- (1)}$$

$$(OO')^2 = O'A^2 = 4^2 + 3^2 = 16 + 9$$

$$= 25 \text{ --- (2)}$$

- From (1) and (2), we have: $OA^2 = (OO')^2 + O'A^2$

i.e., $H^2 = B^2 + P^2$

$\Rightarrow \Delta OO'A$ is a right angled Δ .

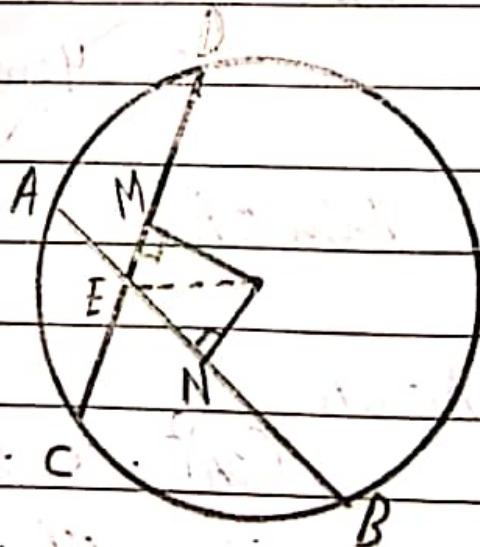
$\Rightarrow OO' \perp AB$

$\Rightarrow BO' = AO' = 3\text{cm}$ ($\because$ $\perp$ from the centre bisects the chord).

$\therefore$ Length of Chord $AB = AO' + BO'$
 $= 3\text{cm} + 3\text{cm}$
 $= 6\text{cm}$

Q2~ If two equal chords of a circle intersect within the circle, ----?

Sol~



- Given- A circle with centre 'O' chords AB and

CD intersect at point E, and $AB = CD$.

- To prove- $ED = EB$
 $CE = AE$.

- Const. 1) Draw $ON \perp AB$, $OM \perp CD$
2) Join EO.

- Proof- In $\triangle EMO$ and $\triangle ENO$.
 $\angle M = \angle N$ (Each 90°)
 $OE = OE$ (Common)
 $OM = ON$ (Chords Equidistant)

- By RHS congruence.

$$\triangle EMO \cong \triangle ENO.$$

$$\Rightarrow EN = EM \text{ (C.P.C.T)} \dots \textcircled{1}$$

- Since $AN = \frac{1}{2}AB \dots \textcircled{2}$ ($\because N$ is mid-point of AB)

$$\text{and } CM = \frac{1}{2}CD.$$

$$\Rightarrow AB = 2AN \dots \textcircled{3}$$

$$\Rightarrow CD = 2CM \dots \textcircled{4}$$

- Since $AB = CD$ (Given)

$$2AN = 2CM \text{ (by 3, 4)}$$

$$\Rightarrow AN = CM \Rightarrow \boxed{CM = AN} \dots \textcircled{5}$$

- Subtracting (1) from (5)

$$CM - EN = AN - EN$$

$$CE = AE$$

$$\Rightarrow \boxed{EC = EA} \text{ --- (6)}$$

- Again $CD = AB$ --- (7) (Given)

$\Rightarrow$ Subtracting (6) from (7)

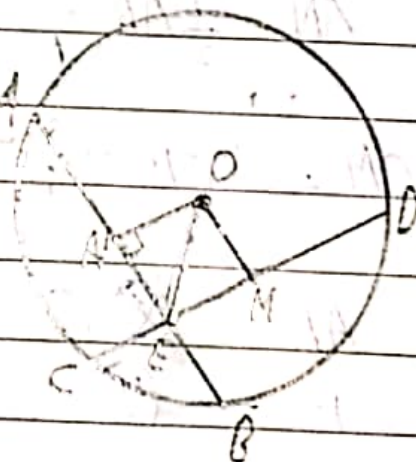
$$CD - EC = AB - EA$$

$$\boxed{ED = EB}$$

Sol
29/08/2026

Q3+ If two equal chords of a circle intersect within the circle, --- ?

Sol



- Given - Two equal chords of a circle intersect

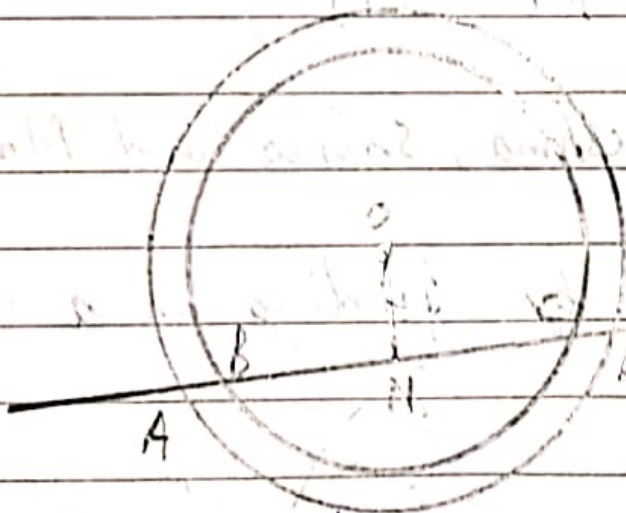
within the circle.

- Proof - In $\triangle OEN$ and $\triangle OEM$
 $OE = OE$ (Common)
 $\angle N = \angle M$ (Each 90°)
 $ON = OM$

- By RHS congruence.
 $\triangle OEN \cong \triangle OEM$
 $\angle 1 = \angle 2$ (C.P.C.T)

Q4) If a line intersects two concentric circles with centre O at A, B, C and D, prove that $AB = CD$.

Sol:-



Two concentric

- Given - Two circles with centre O and point A, B, C and D.

- To Prove - $AB = CD$.

Proof - $ON \perp BC$

$\Rightarrow$ Acc. to Theorem (9.3) i.e; The line drawn from the centre of the circle to the chord bisects the chord.

$$\Rightarrow \cancel{BN} BN = NC \text{ --- (1)}$$

$\Rightarrow$ Now in Bigger circle: $ON \perp AD$

- By Theorem (9.3)

$$\Rightarrow AN = ND \text{ --- (2)}$$

$\Rightarrow$ subtract 1 from 2

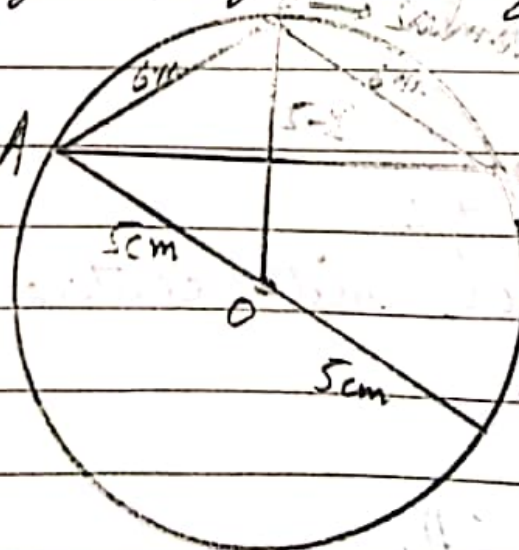
$$\Rightarrow AN - BN = ND - NC$$

$$\boxed{AB = CD}$$

Q5) Three girls Reshma, Salma and Mandip are playing a game by standing on a circle --?

sol)

Reshma - A



Mandip

- Let the position of Reshma, Salma and Mandip be represented by the point A, B and C.

- $AB = BC = 6m$

- Const. Draw $OM \perp BC$
 $OB = 5m$, Let $OM = x$

$\Rightarrow MB = OB - OM$

$MB = 5 - x$

$\Rightarrow$ In right Angled $\triangle AMB$.

- $AB^2 = AM^2 + MB^2$

$6^2 = AM^2 + (5-x)^2 \quad \dots \textcircled{1}$

- Again, In right Angle $\triangle AOM$

$AO^2 = OM^2 + AM^2$

$5^2 = x^2 + AM^2$

$\Rightarrow AM^2 = 5^2 - x^2 \quad \dots \textcircled{2}$

- From 1 and 2, We have.

$6^2 - (5-x)^2 = 5^2 - x^2$

$$36 - (5^2 + x^2 - 2(5)(x)) = 5^2 - x^2$$

$$36 - 5^2 - x^2 + 10x$$

$$10x = 5^2 - x^2 - 36 + 5^2 + x$$

$$10x = 2 \times 5^2 - 36$$

$$10x = 2 \times 25 - 36$$

$$10x = 50 - 36$$

$$10x = 14$$

$$x = \frac{14}{10}$$

$$\boxed{x = \frac{7}{5}}$$

- Using Value of x in (2)

$$AM^2 = 5^2 - x^2$$

$$= 5^2 - \left(\frac{7}{5}\right)^2$$

$$= \left(5 + \frac{7}{5}\right) \left(5 - \frac{7}{5}\right)$$

$$AM^2 = \frac{32}{5} \times \frac{18}{5}$$

$$AM^2 = \frac{576}{25}$$

$$AM = \frac{\sqrt{576}}{25}$$

$$AM = \frac{24}{5}$$

$$AM = 4.8 \text{ m.}$$

⇒ Since M is the mid point of AC. ($\because OM \perp AC$)

$$AC = 2AM$$

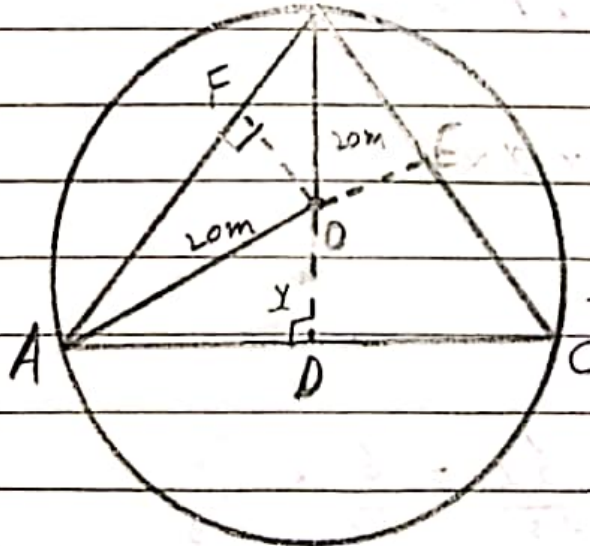
$$AC = 2 \times 4.8$$

$$AC = 9.6 \text{ m}$$

∴ Distance between Salma and Mandip = 9.6 m.

Q6) A Circular park of radius 20m is situated in a colony. The Three boys Ankur, Syed and David are sitting at ^{equal} distance....?

Sol



Let the positions of three boys Ankur, Syed and David be A, B, and C.

- Join OA, OB and OC
- Draw $OD \perp AC$, $OE \perp BC$ and $OF \perp AB$
- Length of $OD = OE = OF = x$
- Since $\triangle ABC$ is an equilateral triangle.
- Let 'a' be the side of the $\triangle ABC$

$$\Rightarrow AB = BC = AC = a$$

$$\text{Area of } \triangle AOC = \text{Area of } \triangle BOC = \text{Area of } \triangle AOB$$

$$\Rightarrow \text{Area of } \triangle ABC = 3 \times \text{Area of } \triangle AOC$$

$$\boxed{\text{Area of equilateral } \triangle = \frac{\sqrt{3}}{4} \cdot \text{side}^2}$$

$$\Rightarrow \frac{\sqrt{3}}{4} a^2 = 3 \times \frac{1}{2} \times AC \times OD$$

$$\frac{\sqrt{3}}{4} a^2 = 3 \times \frac{1}{2} \times a \times x$$

$$\frac{\sqrt{3}}{4} a = \frac{3}{2} \times x$$

$$a = \frac{3x \times 4}{\cancel{2} \sqrt{3}} = \frac{6x}{\sqrt{3}} = \frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} x$$

$$= \frac{a = 2\sqrt{3}x}{2}$$

$$\Rightarrow a = 2\sqrt{3}x \text{ --- ①}$$

- Now in right Angle ΔODC

$$OD^2 + DC^2 = OC^2$$

$$x^2 + \left(\frac{1}{2}AC\right)^2 = 20^2$$

$$- x^2 + \left(\frac{1}{2}2\sqrt{3}x\right)^2 = 400$$

$$= x^2 + 3x^2 = 400$$

$$- 4x^2 = 400$$

$$x^2 = \frac{400}{4}$$

$$x^2 = 100$$

$$x = \sqrt{100}$$

$$\boxed{x = 10}$$

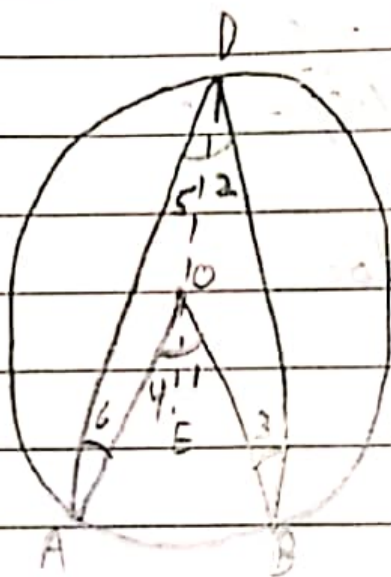
- Using value of x in ①

$$a = 2\sqrt{3} \times 10$$

$$a = 20\sqrt{3} \text{ m}$$

i Length of string = $a = 20\sqrt{3} \text{ m}$

⇒ Theorem 9.7:- The angle subtended by an arc at the centre is double than the angle subtended by it at any point on the remaining part of the circle.



- Given: A circle with centre O and an arc ~~AB~~ AB. D is a point on the remaining part.
- To Prove: $\angle AOB = 2\angle ADB$.
- Const: Join OD and produce it to E.
- Proof: In $\triangle BOD$
 $\angle 1 = \angle 2 + \angle 3$ --- ①
 Since, $OD = OB$ (Radii of same circle).

$$\angle 3 = \angle 2 \dots \textcircled{2}$$

Using $\textcircled{2}$ in $\textcircled{1}$ we have.

$$\angle 1 = \angle 2 + \angle 2$$

$$\angle 1 = 2\angle 2 \dots \textcircled{3}$$

Similarly, $\angle 4 = 2\angle 5 \dots \textcircled{4}$

Adding $\textcircled{3}$ and $\textcircled{4}$, we have.

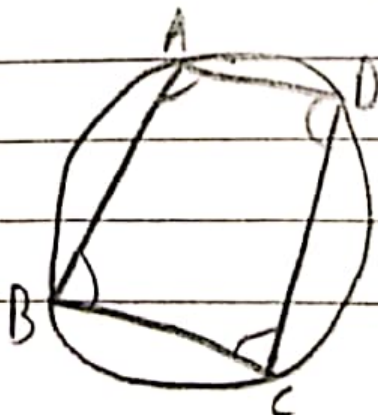
$$\angle 1 + \angle 4 = 2\angle 2 + 2\angle 5$$

$$\angle 1 + \angle 4 = 2(\angle 2 + \angle 5)$$

$$\boxed{\angle AOB = 2\angle ADB}$$

$\Rightarrow$ Cyclic Quadrilateral:-

If the four corners of a Quadrilateral lies on the circumference of a circle, then it is called Cyclic Quadrilateral.

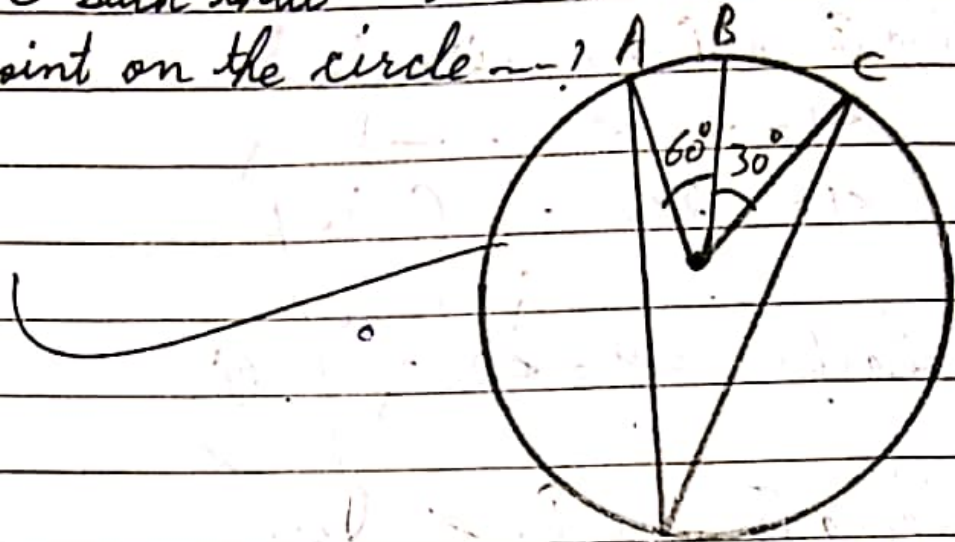


$$\angle A + \angle C = 180^\circ$$

$$\angle B + \angle D = 180^\circ$$

Ex = 8.3

Q1) A, B and C are three points on a circle with centre O such that $\angle BOC = 30^\circ$ and $\angle AOB = 60^\circ$. If D is a point on the circle ---?



Sol: Angle subtended by the arc AC at the centre
 $= 60^\circ + 30^\circ = 90^\circ$

Since, $\angle AOC = 2\angle ADC$

$\rightarrow \angle AOC$

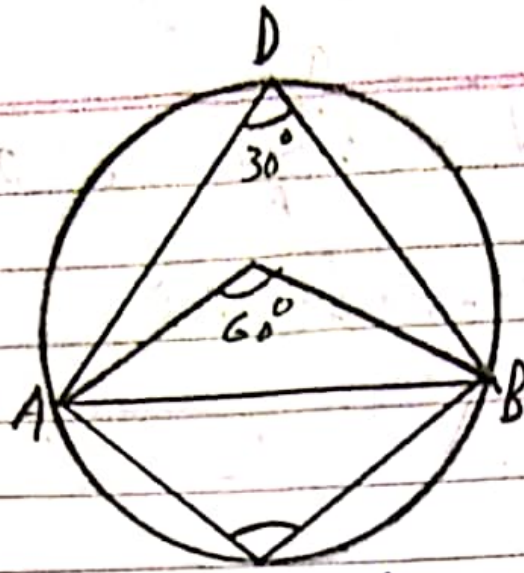
$$90^\circ = 2\angle ADC$$

$$\frac{90^\circ}{2} = \angle ADC$$

$$\Rightarrow \angle ADC = 45^\circ$$

Q2) A chord of a circle is equal to the radius of the circle ---?

Sol)

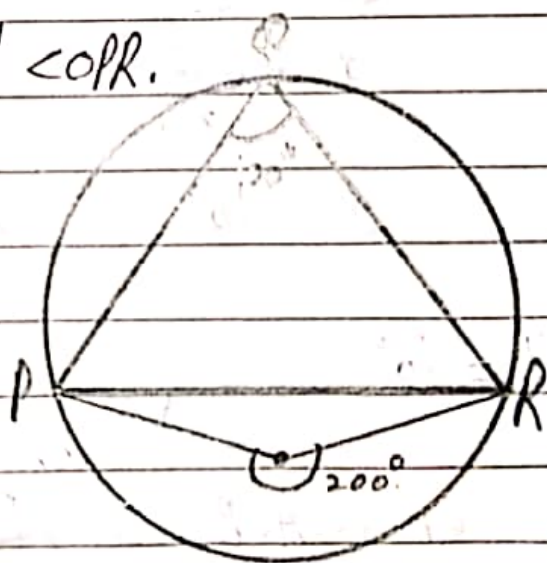


⇒ Since ACBD is a cyclic quadrilateral.

$$\begin{aligned}
 \therefore \angle D + \angle C &= 180^\circ \\
 \angle C &= 180^\circ - \angle D \\
 &= 180^\circ - 30^\circ \\
 &= 150^\circ
 \end{aligned}$$

Q3:- $\angle PQR = 100^\circ$, where P, Q and R are points on a circle with centre O. Find $\angle OPR$.

Sol)



- Since Angle subtended by the chord PR at the pt. Q is 100° .

$\therefore$ Angle subtended at the centre = 200°

$$\begin{aligned}\therefore \angle POR &= 360^\circ - \text{Reflex } \angle POR \\ &= 360^\circ - 200^\circ \\ &= 160^\circ\end{aligned}$$

$$\therefore \boxed{\angle POR = 160^\circ}$$

- In $\triangle OPR$

$OP = OR$ (Radii of same circle)

$$\angle ORP = \angle OPR = x$$

$$\therefore x + x + 160^\circ = 180^\circ$$

$$160^\circ + 2x = 180^\circ$$

~~$$2x = 180^\circ - 160^\circ$$~~

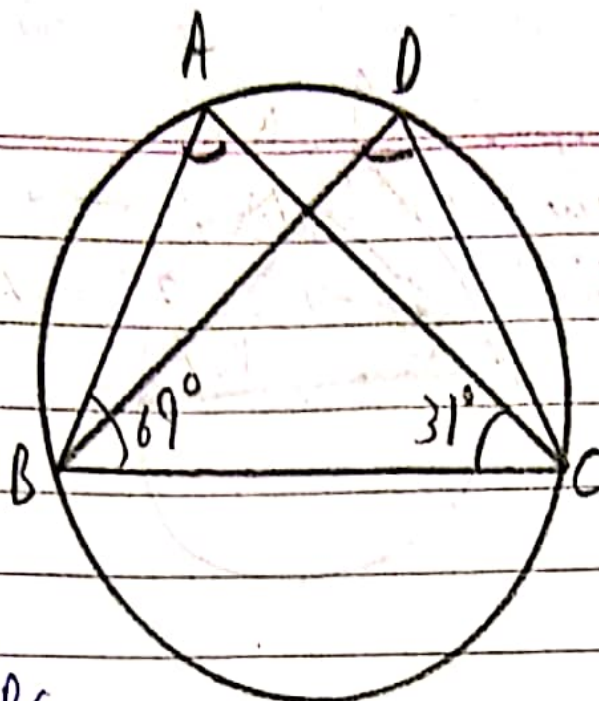
~~$$2x = 20^\circ$$~~

~~$$x = \frac{20^\circ}{2}$$~~

$$x = 10^\circ$$

$$\therefore \boxed{\angle OPR = 10^\circ}$$

Q4) $\angle ABC = 69^\circ$, $\angle ACB = 31^\circ$, Find $\angle BDC$.



In $\triangle ABC$,

$$\angle BAC + \angle ABC + \angle BCD = 180^\circ$$

$$\angle BAC + 69^\circ + 31^\circ = 180^\circ$$

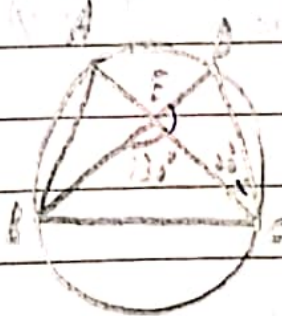
$$\angle BAC + 100^\circ = 180^\circ$$

$$\angle BAC = 180^\circ - 100^\circ$$

$$\angle BAC = 80^\circ$$

$$\therefore \angle BDC = 80^\circ$$

25. A, B, C and D are four points on a circle....?



Sol.)

$$\angle BEC + \angle CED = 180^\circ \text{ (L.D.)}$$

$$130^\circ + \angle CED = 180^\circ$$

$$\angle CED = 180 - 130^\circ$$

$$\angle CED = 50^\circ$$

$$\Rightarrow \angle CED + \angle DCE + \angle EDC = 180^\circ \text{ (Angle Sum)}$$

$$\Rightarrow 50^\circ + 20^\circ + \angle EDC = 180^\circ$$

$$\angle EDC = 180^\circ - 70^\circ$$

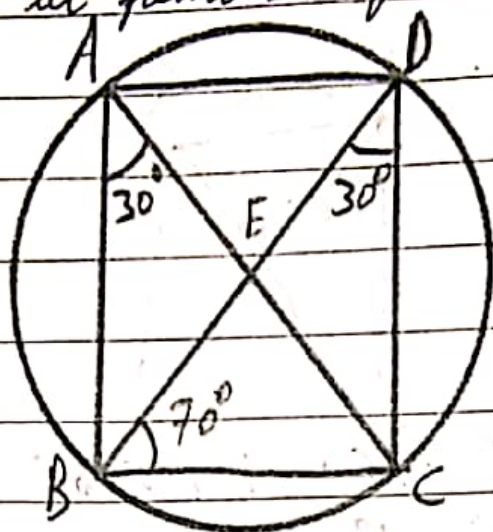
$$\angle EDC = 110^\circ$$

hly,

$\angle BAC = 110^\circ$. (Angle in the same segment)

Q6:- ABCD is a cyclic quadrilateral ~~what~~ whose diagonals intersect at point E. If $\angle DBC = 70^\circ$, $\angle BAC$ is 30° , ...?

Sol



$$\angle BDC = \angle BAC = 30^\circ \text{ (Angles in the same seg.)}$$

- In $\triangle DBC$,

$$30^\circ + 70^\circ + \angle C = 180^\circ$$

$$\Rightarrow 100^\circ + \angle C = 180^\circ$$

$$\angle C = 180^\circ - 100^\circ$$

$$\angle C = 80^\circ \Rightarrow \boxed{\angle BCD = 80^\circ}$$

- If $AB = BC$

$$\angle BCA = \angle BAC = 30^\circ$$

$$\Rightarrow \angle BCA = 30^\circ$$

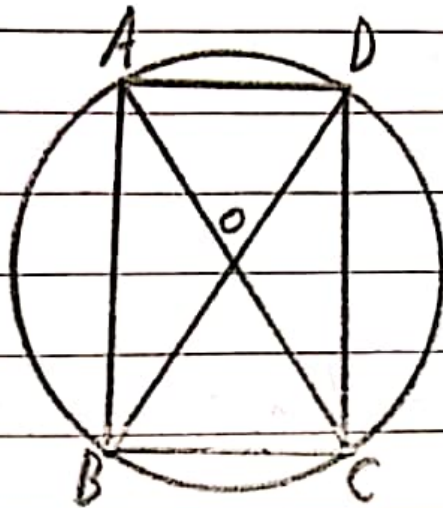
$$\therefore \angle ECD = 80^\circ - 30^\circ = \boxed{50^\circ}$$

Sum
10/09/2016

Ex = 0.3

Q7) If diagonals of a cyclic quadrilateral are diameters of the circle through the vertices of the quadrilateral, prove that it is a rectangle.

sol:



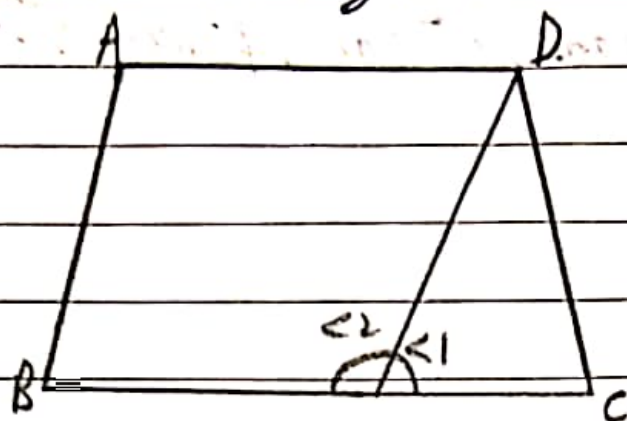
- Taking Diameter
AC (Diagonal of Quadrilateral).

$$\angle ADC = \angle ABC = 90^\circ \quad (\because \text{Angle subtended by diameter AC is } 90^\circ)$$

- Taking BD as diameter.
 $\angle BAD = \angle BCD = 90^\circ$ (Thales Theorem)

$\Rightarrow$ All angles are 90° .
Hence Quad, ABCD is a rectangle.

Q8) If non-parallel sides of a Trapezium are equal, prove that it is cyclic.



sol) Let ABCD be a trapezium such that $AB = CD$.

- Draw $DE \parallel AB$
- $AB \parallel DE$, $AD \parallel BE$ and $AB = DE$, $AD = BE$
- $ABED$ is a $\parallel gm$.
- $\angle A = \angle 2$ --- ①
- Now, $DE = AB = CD$.

$$\Rightarrow DE = CD$$

$\therefore$ In $\triangle DEC$

$$DE = DC$$

$$\angle C = \angle 1$$
 --- ②

$\rightarrow$ Adding ① and ②, we have.

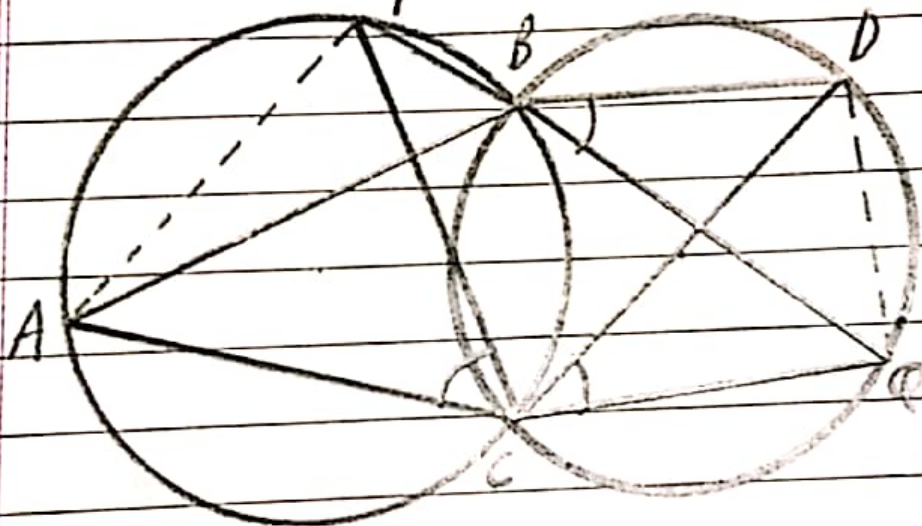
$$\begin{aligned} \angle A + \angle C &= \angle 2 + \angle 1 \\ &= 180^\circ \text{ (L.P.)} \end{aligned}$$

$$\angle A + \angle C = 180^\circ$$

$$\text{Hence, } \angle B + \angle D = 180^\circ$$

$\Rightarrow$ Hence ABCD is Cyclic Trapezium.

Q9) Two circles intersect at two points B and C. Through B, two line segments ABD and PBC are drawn to intersect the circle at A, D and P, Q respectively. Prove that $\angle ACP = \angle QCD$.



$$\angle ACP = \angle ABP \text{ --- (1) (Angles on the same segment)}$$

$$\Rightarrow \angle ABP = \angle QBD \text{ --- (2) (Vertically Opp)}$$

$$\angle QBD = \angle QCD \text{ --- (3) (Same seg.)}$$

- From (1), (2) and (3).

$$\Rightarrow \angle ACP = \angle ABP = \angle QBD = \angle QCD.$$

$$\Rightarrow \boxed{\angle ACP = \angle QCD}$$